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| 2 | <b>Answer: D</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|   | <p>Velocity is a vector, thus the direction sign of the velocity when the ball is going up is the negative of the direction sign of the velocity when the ball is coming back down. Thus, reject option A and option B.</p> <p>Given the asymmetry of the height-time graph where the time taken for upward journey of ball is shorter than downward journey, air resistance must be present. Since work is continually done against air resistance, the final kinetic energy of the ball (just before hitting ground) would be less than the initial kinetic energy of the ball (when it was first thrown up from the ground). Also, the area under graph for the first part of graph should be same as that in the second part since the distance travelled up and down are the same. So, reject option C.</p> |

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| 3 | <b>Answer: C</b> |
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